

ETHOL TABLET 400MG



DESCRIPTION

Ethol Tablet 400mg

A 12 mm diameter, round, convex, scored marked *U.P* white to off white tablet containing Ethambutol Hydrochloride 400mg

INDICATIONS

Ethambutol is indicated in combination with other antituberculosis medications in the treatment of all forms of tuberculosis, including tuberculosis meningitis, caused by *Mycobacterium tuberculosis*.

Ethambutol is used in the treatment of atypical mycobacterial infections such as *Mycobacterium avium* complex (MAC).

PHARMACODYNAMICS

Ethambutol is a synthetic, bacteriostatic antitubercular agent. Its mechanism of action is not fully known. It diffuses into mycobacteria and appears to suppress multiplication by interfering with RNA synthesis. It is effective only against mycobacteria that are actively dividing.

PHARMACOKINETICS

About 80% of an oral dose of ethambutol is absorbed from the gastro-intestinal tract, and the other 20% is excreted unchanged via the faeces. Ethambutol is widely distributed to most tissues and fluids, except cerebrospinal fluid (CSF). CSF concentrations are 10 to 50 % of the corresponding serum concentrations. The concentration in erythrocytes is equal to or double the plasma concentrations, which provide a depot effect for 24 hours. It has been reported to cross the placenta and is distributed into breast milk. Ethambutol does not penetrate intact meninges, but 10 to 50 % may penetrate the meninges of patients with tuberculous meningitis. About 15 % is metabolized in the liver to the aldehyde and dicarboxylic acid derivatives which are inactive and then excreted in the urine. Most of a dose appears in the urine within 24 hours as unchanged drug and 8 to 15 % as the inactive metabolites. The elimination half-life following oral administration in normal renal patients is 3 to 4 hours and in impaired renal function patients it is up to 8 hours. Time taken to reach peak serum concentration is 2 to 4 hours. Peak serum concentration after a single dose of 25 mg per kg of body weight (mg/kg) is 2 to 5 mcg per ml. It has low protein binding of about 20 to 30 %. In dialysis, ethambutol is removed from the blood by haemodialysis and peritoneal dialysis.

DOSAGE

Adult and Children of 13 years of age and over

Tuberculosis: In combination with other antituberculosis medications, 15 to 25 mg per kg of body weight once a day; or 50 mg per kg of body weight, up to 2.5 grams, two times a week; or 25 to 30 mg per kg of body weight, up to 2.5 grams, three times a week.

Adult prescribing limits: Tuberculosis -2.5 grams daily.

Atypical mycobacterial infections: 15 to 25 mg per kg of body weight once a day.

Pediatric

Children up to 13 years of age: Dosage has not been established. However, ethambutol should be considered for all children with organisms resistant to other medications, and in whom susceptibility to ethambutol has been demonstrated or likely. Ethambutol is generally not recommended in children whose visual acuity cannot be monitored (younger than 6 years of age).

SIDE EFFECTS

Incidence less frequent

Acute gouty arthritis (chills; pain and swelling of joints, especially big toe, ankle, or knee; tense, hot skin over affected joints), confusion; disorientation; gastrointestinal disturbances; headache.

Incidence rare

Hypersensitivity (skin rash, fever, joint pain); peripheral neuritis (numbness, tingling, burning pain, or weakness in hands or feet); retrobulbar optic neuritis (blurred vision, eye pain, red-green color blindness, or any loss of vision).

CONTRAINDICATIONS

Ethambutol is contraindicated in patients who are known to be hypersensitive to the drug. It is also contraindicated in patients with optic neuritis.

175mm(w) x 188mm (h)



Black 100%

PRECAUTIONS

It should be given in reduced dosage to patients with impaired kidney function and dosage adjustments may need to be made according to serum concentrations of ethambutol. It should be used with great care in patients with visual defects, the elderly and in children in whom evaluation of changes in visual acuity may be difficult; it should not be used in children under at least 6 years and some consider that it should not be used in children under 13 years of age nor in patients with visual defects. Ocular examination is recommended before treatment with ethambutol. Patients should be advised to report visual disturbances immediately and to discontinue ethambutol pending visual evaluation. Ethambutol may precipitate attacks of gout as it may increase uric acid concentrations.

DRUG INTERACTIONS

Concurrent administration of ethambutol with other neurotoxic medications may increase the potential for neurotoxicity, such as optic and peripheral neuritis.

Laboratory Value Alteration

With physiology / laboratory test values

Uric acid, serum concentrations may be increased.

OVERDOSAGE

No information available.

TOXICOLOGY

Carcinogenicity:

Studies with ethambutol have not been done.

Tumorigenicity/ Mutagenicity/ Pregnancy/ Reproduction:

No information is available.

Fertility:

No information is available.

Pregnancy:

It is recommended that pregnant women with tuberculosis be treated for a minimum of 9 months with multidrug therapy, including ethambutol. Ethambutol crosses the placenta, resulting in fetal plasma concentrations approximately 30% of maternal plasma concentrations. However, problems in humans have not been documented. Studies in mice given high doses have shown that ethambutol causes a low incidence of cleft palate, exencephaly, and vertebral column abnormalities. In addition, studies in rats given high doses have shown that ethambutol may cause abnormalities of cervical vertebrae. Studies in rabbits given high doses have shown that ethambutol may cause monophthalmia, limb reduction defects, hare lip and cleft palate.

Breast-feeding:

Ethambutol is distributed into breast milk in concentrations approximating maternal serum concentrations. However, problems in humans have not been documented.

Pediatrics:

Appropriate studies on the relationship of age to the effects of ethambutol have not been performed in children up to 13 years of age. Ethambutol is generally not recommended in children whose visual acuity cannot be monitored (younger than 6 years of age). However, ethambutol should be considered for all children with organisms resistant to other medications and in whom susceptibility to ethambutol has been demonstrated or is likely.

Geriatrics:

No information is available on the relationship of age to the effects of ethambutol in geriatric patients. However, elderly patients are more likely to have age-related decrease in renal function, which may require an adjustment of dosage in patients receiving ethambutol.

PRESENTATION

400mg : 50 x 10's (Blister pack)

STORAGE CONDITIONS AND USER INSTRUCTIONS

Do not stop taking this medicine except on your doctor's advice.

Store in a dry place below 30°C.

Protect from light.

Keep out of reach of children.

Jauhi daripada kanak-kanak.

Shelf life: please refer to outer package.

Route of administration: Oral

Product Registration Holder:

Duopharma Manufacturing (Bangi) Sdn. Bhd.

Lot No. 2, 4, 6, 8 & 10, Jalan P/7, Section 13,
Bangi Industrial Estate, 43650 Bandar Baru Bangi,
Selangor, Malaysia.

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